# XIAOYUN XU

xuxiaoyun@bza.edu.cn ⋄ https://xiaoyunxxy.github.io/

## ACADEMIC APPOINTMENT

**Zhongguancun Academy, Beijing** July, 2026 - Current
Assistant Professor, topic: AI Security

**Radboud University, Nijmegen, The Netherlands** Nov, 2025 - July, 2026
Postdoc, topic: Adversarial Machine Learning

## EDUCATION

**Radboud University, Nijmegen, The Netherlands** May, 2022 - Jan, 2026
PhD in Computer Science
Supervisor: Dr. Stjepan Picek
Thesis: No Time to Spare: Adversarial Machine Learning at Training and Inference Time

**University of Bristol, Bristol, UK** 2017 - 2018
MSc in Advanced Computing
Thesis: Investigating the effectiveness of existing machine-learning-based compiler optimization techniques

**University of Electronic Science and Technology of China** 2013 - 2017
BEng in Software Engineering
Thesis: Vehicle license plate recognition based on SVM and ANN

## RESEARCH PUBLICATION

1. Behrad Tajalli, Henk Berendsen, **Xiaoyun Xu***, Stjepan Picek. Quiet Triggers, Loud Footprints: A Tri-Space Measurement Study of Backdoor Stealthiness. *IEEE Symposium on Security and Privacy (SP)*, 2027.
2. **Xiaoyun Xu**, Shujian Yu, Zhuoran Liu, and Stjepan Picek. MIMIR: Masked Image Modeling for Mutual Information-based Adversarial Robustness. *The Network and Distributed System Security (NDSS)*, 2026.
3. Siyu Zhang, Yelu Gu, **Xiaoyun Xu***, Kirsten Cater, Oussama Metatla. AdaptaFace: Adversarial Assistance for Inclusive Facial Gesture Interaction. *ACM Symposium on User Interface Software and Technology (UIST)*, 2026.
4. Siyu Zhang, Wenda Zhao, Yelu Gu, Fanzhaoyang Lei, **Xiaoyun Xu**, Anne Roudaut, Oussama Metatla. Chameleon: Negotiating Machine Visibility with Programmable Privacy Surfaces (Poster). *ACM Symposium on User Interface Software and Technology (UIST)*, 2026.
5. Siyu Zhang, Wenda Zhao, Yelu Gu, Fanzhaoyang Lei, **Xiaoyun Xu**, Oussama Metatla. Position: Persistent AI Agents Require Representational Consent Beyond Memory Permission (Short Paper). *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp)*, 2026.
6. **Xiaoyun Xu**, Zhuoran Liu, Stefanos Koffas, and Stjepan Picek. Towards Backdoor Stealthiness in Model Parameter Space. *ACM Conference on Computer and Communications Security (CCS)*, 2025.
7. **Xiaoyun Xu**, Zhuoran Liu, Stefanos Koffas, Shujian Yu, and Stjepan Picek. BAN: Detecting Backdoors Activated by Adversarial Neuron Noise. *Advances in Neural Information Processing Systems (NeurIPS)* , 2024.
8. Zhuoran Liu, Senna van Hoek, Péter Horváth, Dirk Lauret, **Xiaoyun Xu**, and Lejla Batina. Real-world Edge Neural Network Implementations Leak Private Interactions Through Physical Side Channel. *arXiv preprint*, 2025.
9. **Xiaoyun Xu**, Oguzhan Ersoy, Behrad Tajalli, and Stjepan Picek. Universal Soldier: Using universal adversarial perturbations for detecting backdoor attacks. *IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, 2024.

10. **Xiaoyun Xu**, and Stjepan Picek. Poster: Boosting Adversarial Robustness by Adversarial Pre-training. *ACM Conference on Computer and Communications Security (CCS)*, 2023.
11. **Xiaoyun Xu**, Guilherme Perin, and Stjepan Picek. IB-RAR: Information Bottleneck as Regularizer for Adversarial Robustness. *IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, 2023.
12. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, Qianru Meng, Weiheng Li, and Yanjun Wu. AI-CTO: Knowledge graph for automated and dependable software stack solution. *Journal of Intelligent and Fuzzy Systems*, 2021.
13. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, Xu Duan, Weiheng Li, Yanjun Wu, and Bin Wu. Information leakage by model weights on federated learning. *In Proceedings of the 2020 workshop on privacy-preserving machine learning in practice, CCS workshop PPLMP*, 2020.
14. **Xiaoyun Xu**, Jingzheng Wu, Mutian Yang, Tianyue Luo, A method of shortening vulnerability attack window based on knowledge graph reasoning, CN110378126B, (Patent)
15. Vulnerability management platform based on Knowledge Graph, 2019SR0860641, (Software Copyright)

## RESEARCH EXPERIENCE AND PROJECTS

**Horizon Europe – the Framework Programme for Research and Innovation** Nov, 2025 - Present

- Project Title: Secure Hardware and Software for AI systems (SHASAI)
- Grant agreement ID: 101225866

**Dutch national e-infrastructure (SURF Cooperative)** Jan, 2025 - Dec, 2025

- Project Title: LLM pruning against backdoor attacks
- grant no. EINF-12068

**Dutch national e-infrastructure (SURF Cooperative)** Sep, 2024 - Sep, 2025

- Project Title: Adversarially train the projection layer for robustness of vision and language alignment
- grant no. EINF-10853

**Institute of Software, Chinese Academy of Sciences** 2018 - 2020

- Research Assistant, topics: Knowledge Graph, security vulnerabilities, and Interpretable AI
- Domain knowledge graph of software vulnerabilities

## ACADEMIC SERVICES

| | |
|---|---|
| **Reviewers/PC** | TIFS, BMVC, ICLR, NeurIPS, USENIX Security, SaTML, CCS |
| **External Reviewers** | S&P, NDSS |

## TEACHING

- Guest Lecturer, Academic Writing and Research Methods (NWI-IBC043), 2025, Radboud University
- Teaching Assistant, Security and Privacy of Machine Learning (NWI-IMC069), 2023, Radboud University

## CAMPUS EXPERIENCE

**Summer School, Šibenik, Croatia** 2022.06

- Topic: On real-world crypto and privacy

**Exchange program at Christopher Newport University, America** 2015.07 – 2015.08

- Received 10,000 (Chinese yuan) scholarship
- Explored the American education model and culture

**Exchange program at Waseda University, Japan** 2016.01 – 2016.02

- Received a scholarship of 50,000 yen (JPY)

- Learned more about Japanese culture and visited Japanese well-known enterprises

## SKILLS

| | |
|---|---|
| **Languages** | Chinese, English. |
| **Programming** | C/C++, Python, Linux, HTML, Git, CSS, Makefile, NodeJS, JavaScript, Neo4j, Cypher. |
| **Courses** | Computational Neuroscience, Cloud Computing, Computer Graphics, Data Structures and Algorithms, Database Principles and Applications, Computer Networks, Image Processing, Computer Vision, etc. |